Data Structures and Algorithms Laboratory

Assignment 1 (2 Weeks)

1. Define an ADT List, L with N elements, which are stored in a file. The associated operations are given below. Write a menu driven program for the same using (i) Array and (ii) Linked list.
   a) Create (L): Create a list of N elements.
   b) Length (L): Returns the length of the list L, N.
   c) PrintList (L): Print the elements of L in order of occurrence left-to-right (or right-to-left).
   d) Retrieve (i, L): Retrieve the ^{i}th element of L.
   e) Store (x, i, L): Store a new value, x at ^{i}th position in L.
   f) Insert (x, i, L): Insert a new element at ^{i}th position of L, moving the elements at i and the following positions to the next higher position.
   g) Delete (i, L): Delete the ^{i}th element from L, moving the elements of the following positions to the next lower position.
   h) Search (x, L): Search the key x in the list L. If found, return its position in the list, otherwise return (-1) indicating not found.
2. Given two sorted lists, L_1 and L_2. Write a function to compute $L_1 \cap L_2$ using only the basic list operations.
3. Given two sorted lists, L_1 and L_2. Write a function to compute $L_1 \cup L_2$ using only the basic list operations.
4. Write a function to add two polynomials.
5. Write a program to evaluate a postfix expression.
6. Write a program to convert a postfix expression to infix expression.
7. Write a program to implement three stacks in one array. Perform stack operations by taking the elements from a file. The selection of the stack is to be done randomly.
8. Write the functions to implement queue using linked list. Take the elements from a file, which is to be given as the command line argument. Perform insertion and deletion randomly.